

OLOVYANNAYA, Russian Federation

WMO: 309610

Lat: 50.950N

Lon: 115.583E

Elev: 2098

StdP: 13.62

Time Zone: 9.00 (E09)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.4	-26.7	-35.7	0.8	-30.8	-31.2	1.0	-26.0	26.3	-0.4	21.0	-3.1	2.3	150	0.650

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	88.8	65.5	84.7	64.4	81.0	63.3	69.7	81.2	67.6	79.0	65.8	76.1	9.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	103.2	75.2	63.9	96.0	72.5	62.1	90.3	70.7	35.2	80.9	33.4	79.6	31.9	76.0	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	24.6	20.9	17.9	DB	-37.8	95.8	5.8	5.4	-42.0	99.8	-45.5	102.9	-48.7	106.0	-53.0	109.9
(5)				WB	-37.8	74.3	5.8	5.6	-42.0	78.4	-45.4	81.7	-48.7	84.9	-52.9	89.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	32.3	-10.3	-1.6	16.9	37.6	52.2	64.6	69.5	64.4	52.0	34.6	11.3	-6.1	(6)
(7)		DBStd	29.42	9.25	9.86	11.96	9.13	8.49	7.43	5.64	6.39	8.17	8.90	11.70	10.22	(7)
(8)		HDD50	8253	1869	1444	1026	385	73	1	0	1	71	483	1162	1738	(8)
(9)		HDD65	12271	2334	1864	1491	822	406	94	18	88	396	943	1612	2203	(9)
(10)		CDD50	1781	0	0	0	13	142	440	603	447	131	5	0	0	(10)
(11)		CDD65	324	0	0	0	0	10	83	156	69	6	0	0	0	(11)
(12)		CDH74	3149	0	0	0	6	159	920	1402	580	82	0	0	0	(12)
(13)		CDH80	1053	0	0	0	2	46	334	488	166	17	0	0	0	(13)
(14)	Wind	WSAvg	7.1	5.1	5.7	7.4	9.7	10.3	7.6	6.6	6.8	7.4	6.4	5.0	(14)	
(15)	Precipitation	PrecAvg	13.2	0.2	0.1	0.2	0.5	0.9	2.5	3.3	2.9	1.3	0.4	0.4	0.2	(15)
(16)		PrecMax	22.6	0.5	0.5	0.7	1.6	2.2	7.3	10.1	6.1	3.9	1.0	1.1	0.6	(16)
(17)		PrecMin	7.3	0.0	0.0	0.0	0.0	0.1	0.2	1.1	0.8	0.2	0.0	0.0	0.0	(17)
(18)		PrecStd	3.9	0.1	0.1	0.2	0.5	0.6	1.9	2.0	1.6	0.9	0.3	0.2	0.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.4	30.0	52.8	71.4	85.9	96.5	94.7	90.2	82.1	64.9	43.0	23.9	(19)
MCWB			14.8	24.4	39.0	49.0	58.1	67.6	68.8	67.1	60.7	49.0	35.1	20.7	(20)	
2%		DB	11.3	23.4	45.1	63.6	78.0	88.0	89.6	84.6	75.4	57.8	36.2	16.3	(21)	
		MCWB	9.7	19.8	34.2	46.2	55.7	63.0	67.8	65.4	58.9	44.6	30.1	14.3	(22)	
5%		DB	6.6	18.4	39.9	57.8	72.8	83.1	85.2	80.3	70.2	52.5	32.0	11.6	(23)	
		MCWB	5.2	15.7	31.1	42.2	53.5	62.7	65.6	64.0	56.2	41.4	27.6	10.0	(24)	
10%		DB	2.7	14.0	34.6	52.8	68.1	78.9	81.4	76.3	66.0	48.0	27.5	7.6	(25)	
		MCWB	1.5	11.9	27.6	39.6	51.1	61.4	64.9	62.7	54.3	38.9	23.8	6.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.9	24.8	39.6	50.4	61.1	72.2	74.8	71.2	63.9	49.9	35.5	20.7	(27)
MCD B			17.2	29.6	51.2	68.3	75.5	86.7	85.9	81.1	76.7	63.5	42.0	23.6	(28)	
2%		WB	9.7	19.7	34.7	46.7	57.6	67.7	70.5	67.7	60.2	45.7	30.6	14.3	(29)	
		MCD B	11.2	23.1	44.2	62.4	74.6	81.3	81.5	78.7	70.9	56.1	35.6	16.1	(30)	
5%		WB	5.3	15.8	31.2	43.1	54.8	65.0	68.4	65.8	57.9	42.3	27.6	10.0	(31)	
		MCD B	6.6	18.4	39.3	56.4	70.4	77.5	79.8	75.5	67.7	51.7	32.0	11.4	(32)	
10%		WB	1.5	12.1	27.9	40.0	52.1	62.9	66.9	64.3	55.5	39.4	24.0	6.4	(33)	
		MCD B	2.6	13.9	34.2	52.2	66.5	75.2	77.5	73.3	64.3	47.3	27.4	7.5	(34)	
(35)	Mean Daily Temperature Range	MDBR	15.4	19.2	20.0	19.9	21.8	20.2	18.7	18.4	20.1	18.2	16.2	14.8	(35)	
5% DB		MCD BR	17.5	21.8	23.5	27.3	28.9	26.6	25.1	25.1	26.6	24.5	19.4	17.1	(36)	
		MCWBR	16.0	18.9	15.4	15.6	13.8	11.2	9.6	10.8	13.6	14.8	14.6	15.5	(37)	
		MCD BR	17.8	21.9	22.9	26.1	26.3	21.9	20.3	19.1	23.0	23.0	19.4	17.0	(38)	
5% WB		MCWBR	16.3	19.0	15.6	15.6	14.1	11.5	10.2	10.2	13.2	14.7	14.8	15.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.250	0.278	0.317	0.411	0.486	0.462	0.420	0.387	0.342	0.307	0.277	0.257	(40)	
(41)		taud	2.228	2.182	2.169	2.040	1.883	2.072	2.265	2.336	2.398	2.400	2.288	2.189	(41)	
(42)		Ebn at Noon	248	269	278	261	245	253	264	267	269	258	233	221	(42)	
(43)		Edh at Noon	22	30	37	48	59	49	40	36	30	24	21	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	499	869	1360	1682	1906	2004	1843	1617	1285	867	504	376	(44)	
(45)		RadStd	21	41	64	112	112	159	128	107	93	47	22	17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page